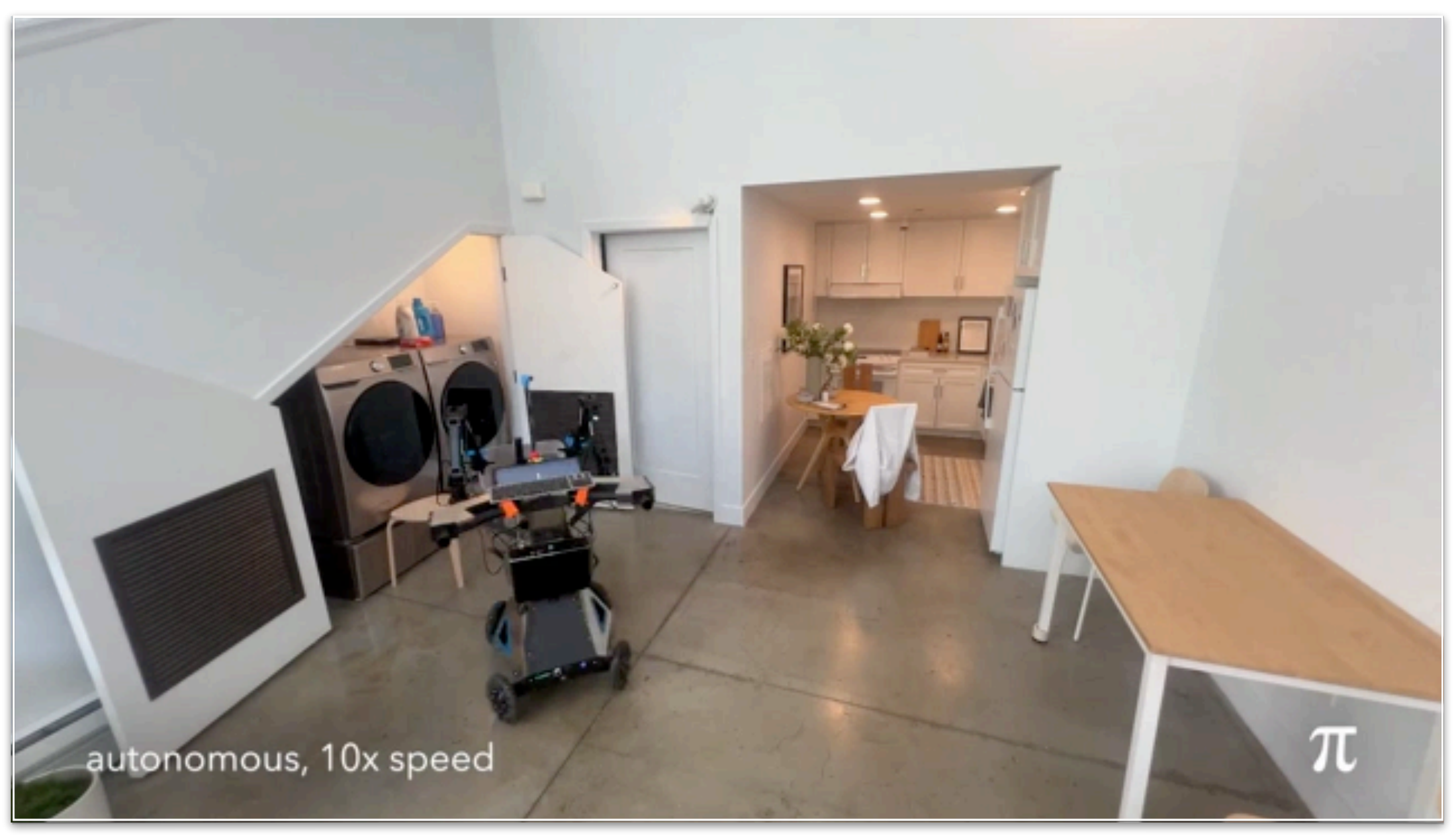


Language as an instructional modality

• Foundation models have popularized natural language as an instructional modality

Greatness is challenging due to the inherent ambiguity of language

A Pi 3 mobile robot, a small black and blue vehicle with four wheels, is positioned in the center of a room. It is equipped with a laptop on top, which is connected to a camera and other sensors. The robot is moving towards a doorway that leads to a kitchen. In the background, there is a dining area with a round wooden table and chairs, and a kitchen with white cabinets and a refrigerator. To the left, there is a laundry room with a washing machine and dryer. The room has a sloped ceiling and a concrete floor. The text "autonomous, 10x speed" is overlaid in the bottom left corner, and the Pi logo is in the bottom right corner.

autonomous, 10x speed

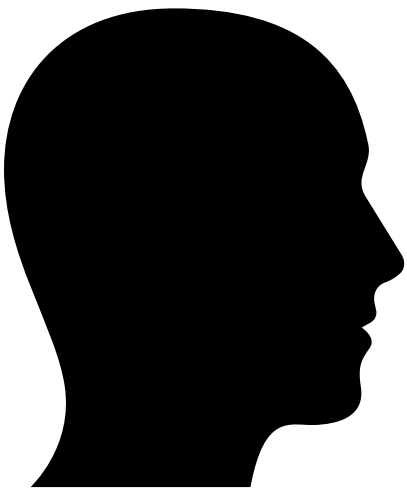
π

Black, Kevin, et al.' π_0 : A Vision-Language-Action Flow Model for General Robot Control.' 2024.

Roccamonde, Juan, et al. "Vision-Language Models are Zero-Shot Reward Models for Reinforcement Learning." ICLR 2024.

Brohan, Anthony, et al. "Rt-1: Robotics transformer for real-world control at scale." 2022.

Sontakke, Sumedh, et al. "Roboclip: One demonstration is enough to learn robot policies." NeurIPS 2023.



**Pick up the laundry
and fold it. Avoid
breaking anything!**







autonomous, 10x speed

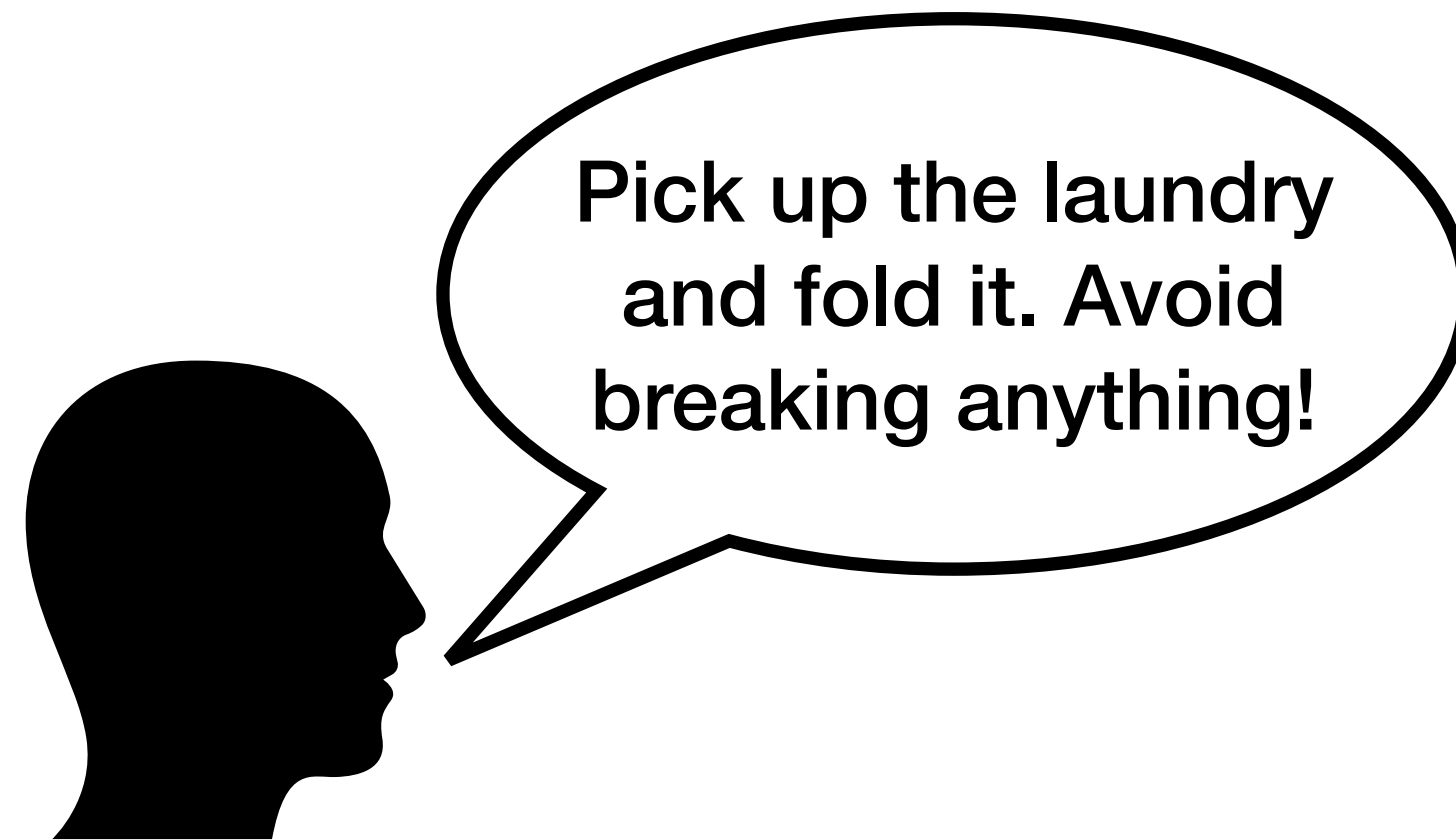
π

A first-person view from a robotic car moving through a modern, minimalist house. The car is a small, black, four-wheeled robot with a laptop and other sensors mounted on top. It is moving on a light-colored concrete floor. To the left, there is a white wall with a large, dark, rectangular vent. Behind the vent, a white door is open, revealing a laundry room with a washing machine and dryer. To the right, there is a white wall and a wooden table. In the background, a kitchen area is visible through an open doorway, featuring white cabinets, a refrigerator, and a dining table with a white cloth. The overall atmosphere is clean and modern.

autonomous, 10x speed

π

Language as an instruction modality



Black, Kevin, et al. " π_0 : A Vision-Language-Action Flow Model for General Robot Control." 2024.

- Foundation models have popularized **natural language** as an instruction modality.
- Correctness is challenging due to the inherent **ambiguity** of language.

Rocamonde, Juan, et al. "Vision-Language Models are Zero-Shot Reward Models for Reinforcement Learning." ICLR 2024.

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Automata as an instruction modality

